

Justin Varghese

Computer Science and Engineering Student

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Professional Summary

Final-year Computer Science student who built a real-time ride-hailing backend with Node.js and Socket.IO, achieving sub-second event dispatch across WebSocket channels. Experienced across the full stack—React and Flutter frontends, Express.js REST APIs, and Python data pipelines—with hands-on integration of OpenAI and Gemini APIs for AI-driven applications. Seeking Full Stack or Backend Engineering roles focused on building scalable, real-time systems.

Technical Skills

- **Languages:** Java, Python, JavaScript, Kotlin, Dart
- **Frameworks & Libraries:** Node.js, Express.js, React (Vite), Flutter, Jetpack Compose, Socket.IO, Pydantic, NetworkX, Matplotlib
- **Databases & Tools:** MySQL, Supabase (PostgreSQL), SQLite, Room DB, Git, GitHub, Postman, REST APIs, WebSockets, OSRM
- **AI & APIs:** OpenAI API (GPT-4o-mini), Gemini API (gemini-2.0-flash), GitHub REST API
- **Core Strengths:** Data Structures & Algorithms, Object-Oriented Programming, Backend Development, API Design, Real-Time Systems, Offline-First Architecture

Professional Experience

Software Development Intern | *Nxtlogic Software Solutions, Coimbatore*

June 2025 – July 2025

- Developed CampusHub, a full-stack campus management application using Node.js, Express.js, MySQL, and JavaScript, implementing role-based authentication across three distinct user roles.
- Designed and integrated RESTful APIs spanning three modules—event management, club management, and lost-and-found—supporting CRUD operations, search, pagination, and server-side input validation.
- Built and optimized a multi-table MySQL schema with foreign key constraints and parameterized queries to enforce referential integrity across all application modules.
- Performed end-to-end API testing with Postman and managed source control using Git and GitHub throughout the development lifecycle.

Projects

Equinox | Ride-Hailing Ecosystem | GitHub

Node.js | Socket.IO | Flutter | Supabase (PostgreSQL) | OSRM

- Engineered an event-driven backend using Node.js and Socket.IO, implementing WebSocket-based bidirectional channels for real-time rider-driver dispatch, ride acceptance, and state management with sub-second event latency.
- Integrated the open-source OSRM routing engine into the backend REST API, computing optimized driving routes with distance and duration estimates while eliminating dependency on paid third-party routing services.
- Developed a cross-platform Flutter mobile application featuring Google Maps integration, real-time GPS tracking via Geolocator, and Provider-based state management for live ride status updates.
- Implemented secure user authentication using Supabase Auth in the Flutter client, supporting sign-up, login, and session management workflows.
- Architected persistent ride lifecycle management using Supabase (PostgreSQL), implementing CRUD operations for ride request, acceptance, location tracking, and completion workflows.

Developer Knowledge Engine | GitHub

Python | OpenAI API (GPT-4o-mini) | NetworkX | SQLite | GitHub REST API | Pydantic

- Built a modular Python ETL pipeline utilizing the GitHub REST API to programmatically fetch, parse, and standardize raw markdown documentation from multiple public repositories.
- Integrated the OpenAI API (GPT-4o-mini) to perform automated entity and relationship extraction from parsed documentation, outputting structured JSON validated with Pydantic schemas.
- Constructed a knowledge graph using NetworkX (DiGraph), mapping structured entity-relationship data into a queryable SQLite database, and visualized the graph topology using Matplotlib.
- Validated end-to-end pipeline integrity through unit and integration tests across the parser, extractor, and pipeline modules.

RepoRoast | AI Code Review System | GitHub

React (Vite) | Node.js | Express.js | Gemini API (gemini-2.0-flash) | GitHub API

- Built and deployed a full-stack code analysis platform on Vercel, combining a React (Vite) single-page application with a Node.js/Express.js REST API backend.

- Integrated the Google Gemini API (gemini-2.0-flash) to perform automated, context-aware code quality reviews on repository-scale codebases.
- Engineered a dynamic repository ingestion pipeline using the GitHub API to recursively fetch file trees and submit source code for AI-driven analysis.
- Implemented intelligent payload optimization—filtering binary files, build artifacts, node_modules/ directories, and files exceeding 100KB—to minimize API token consumption per review request.

Intent Pay | Expense Management App | GitHub
Kotlin | Jetpack Compose | Room DB | Material 3

- Engineered a native Android BroadcastReceiver to intercept incoming bank SMS alerts in real-time, extracting transaction amounts and merchant details using regex-based parsing in under 500ms.
- Built a rule-based transaction classification engine backed by Room DB, categorizing 100% of intercepted transactions into spending categories (Shopping, Transport, Food) entirely on-device with zero cloud dependency.
- Developed a “Reflection” overlay screen using Jetpack Compose that triggers on transactions exceeding a configurable threshold, enforcing a mandatory countdown pause to reduce impulse purchases.
- Designed the full application as offline-first with no network permissions, ensuring complete data privacy and local-only processing using Jetpack Navigation and Material 3.

Volunteer Experience & Leadership

- **Student Coordinator, KANAM Fest** 2025 – 2026
Led the Paper Presentation and Project Expo tracks for KANAM 2025, coordinating end-to-end logistics. Returned as coordinator for KANAM 2026 Project Expo, managing operations for over 150 participants across both editions.
- **NSS Volunteer** 2023 – Present
- **School Pupil Leader** 2022
- **JRC Captain** 2019 – 2022

Education

- **B.E. Computer Science and Engineering**, Dr. N.G.P Institute of Technology Expected March 2027
CGPA: 8.6 (up to 6th Semester)
- **HSC (12th Grade)**, Sacred Heart M.H.S.S, Kayyunni | 83% March 2023
- **SSLC (10th Grade)**, Sacred Heart M.H.S.S, Kayyunni | Pass March 2021

Achievements & Personal Details

- **Hackathons:** Finalist, CODEZAP 2026 Hackathon (final round pending); completed 24-hour Hackathon at KPR Institute and 36-hour Cybersecurity Hackathon at K.S. Rangasamy College.
- **Awards:** Winner, Intra-college Idea Presentation Competition; Technical Paper Presenter at Hindusthan College of Engineering.
- **Languages:** English (Professional), Malayalam (Native), Hindi, Tamil.